DRILL 22

VEHICLE EXTRICATION

NFPA 1001 Level 2 (JPR 5.4.1) / Rural Rescue

VEHICLE STABILIZATION & GLASS MANAGEMENT

⏱ **Time:** 45–60 min | 🚒 **Apparatus:** Rescue / Engine

🦺 **PPE:** Extrication Gloves, Bunker Gear, Eye & Respiratory Protection

👥 **Roles:** Drill Leader (Facilitator) | Safety Officer (Watcher) | Stabilization & Glass Crews

🎯 OBJECTIVE & RURAL FIREGROUND VALUE

Every firefighter must be able to stabilize a vehicle on its wheels or side using **Step Chocks**, **Box Cribbing**, and **Rescue Struts**, chock the wheels to prevent roll, and safely manage **Tempered vs Laminated Glass** without spraying glass fragments over trapped patients. Unstable vehicles and uncontrolled glass cause severe responder and patient injuries at rural highway collisions.

📦 TOOLS & APPARATUS SETUP

- Training vehicle (scrap car, municipal pickup, or mock frame).
- 4 step chocks, 12–16 pieces of 4x4 box cribbing, 4 wedges.
- Mechanical or ratchet rescue struts (if equipped).
- Glass tools: Spring-loaded center punch, glass master / reciprocating saw, heavy duct tape or contact paper, canvas tarp / patient blanket.

⚠️ SAFETY WATCH-OUTS & STOP RULE

⚠️ **Vehicle Suspension Movement:** Never place hands, arms, or boots underneath a vehicle's rocker panels or tires until stabilization is completed.

⚠️ **Glass Dust Inhalation:** Modern windshields (laminated glass) produce fine, hazardous glass dust when cut. Rescuers and patients must wear N95/particulate masks and safety glasses.

🛑 **THE STOP RULE: Shout "STOP!" immediately if the vehicle rocks, shifts, or slips from cribbing while personnel are working around it.**

PHASE 1: 4-POINT VEHICLE STABILIZATION (WHEELS ON GROUND) (15 MINUTES) — SLOW DRY RUN

1. **Prevent Roll:** Place rubber or wooden wheel chocks tightly in front of and behind the uninjured rear tire.
2. **Defeat the Suspension (Step Chocking):**
   - A car resting solely on its inflated tires and coil springs will bounce violently during extrication tool use.
   - Slide **Step Chocks** beneath the rocker panels at four structural points:
   - Just behind the front wheels (front A-pillar).
   - Just ahead of the rear wheels (rear C-pillar).
   - Insert chocks with the steps facing upward; tap wooden wedges into any gaps until the car frame sits solid on the wood.
   - Deflate tires (pull valve stems) to settle the chassis firmly onto the step chocks.
   - Rock test: Push firmly on the vehicle fender. The vehicle should feel rigid with **zero suspension bounce**.

DRILL 22: PRACTICAL EVOLUTIONS & TAILGATE DEBRIEF

Vehicle Extrication • Practical Application

PHASE 2: BOX CRIBBING (VEHICLE ON SIDE / LIFTED LOAD) (15 MINUTES) — BUDDY COACHING

1. **The 4x4 Box Crib:**
   - Build a stable box using two parallel 4x4 timbers, overlaid perpendicularly with two more 4x4s.
   - Rule of thumb: Crib height must never exceed **3 times the contact width** (e.g., an 18" wide base can only be stacked up to 54" high).
   - Top off with angled wedges to ensure 100% surface contact with the angled frame.

PHASE 3: GLASS MANAGEMENT (TEMPERED VS LAMINATED) (20 MINUTES) — TIMED / STRESS

1. **Identify the Glass:**
   - **Tempered Glass (Side & Rear Windows):** Shatters into thousands of small, blunt cubes when struck.
   - **Laminated Glass (Windshields & Many Modern Side Windows):** Two layers of glass bonded to a plastic sheet; will not shatter with a center punch. Look for the "Laminated" bug/stamp in the corner!
2. **Managing Tempered Glass (Side Windows):**
   - Cover window with adhesive film or cross-strips of heavy duct tape to capture fragments.
   - Cover interior patient with a heavy blanket or salvage tarp.
   - Position a **Spring-Loaded Center Punch** in the **lowest, outermost corner** of the window (never center of glass!).
   - Push until punch clicks. Window shatters into tape. Rake glass fragments outward using a gloved hand or tool.
3. **Managing Laminated Glass (Windshield):**
   - Cut around the perimeter using a manual **Glass Master** or reciprocating saw (fine metal blade, 14–18 TPI).
   - Spray water or shaving cream along the cut line to capture glass dust.
   - Peel the cut windshield outward and discard under the vehicle or in the scrap pile.

⚠️ COMMON PITFALLS & FIREGROUND HAZARDS

- ❌ *Punching tempered glass in the center:* The center of the glass flexes and resists the punch. Always punch the lower corner where glass is stiffest!
- ❌ *Stacking cribbing too high:* Tapering or building a narrow, wobbly tower will kick out when hydraulic spreaders exert 10,000+ lbs of force on the door!
- ❌ *Ignoring patient protection:* Shattering a side window without shielding the driver's face sprays sharp cubes directly into their eyes.

🗣️ 5-MINUTE TAILGATE DEBRIEF

1. ***"Did the vehicle bounce after chocks were set?", "How did we distinguish between tempered and laminated side glass?"***

✅ APPARATUS RESET CHECKLIST

- ☐ Glass shards swept clear of training area.
- ☐ Cribbing and step chocks inspected for splits or rot; stacked neatly in truck trays.
- ☐ Center punches and saw blades cleaned and returned to rescue box.
- ☐ Attendance roster signed.

DRILL LEADER SIGNATURE	SAFETY OFFICER SIGNATURE	DATE COMPLETED	TRAINING HOURS